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DAMFusion: An end-to-end degradation-aware multi-modal image fusion framework

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ABSTRACT

Multi-modal image fusion aims to integrate complementary information from different modalities into a more comprehensive and robust unified representation, thereby better supporting downstream tasks such as object detection and image recognition. However, most existing methods often lack explicit modeling capabilities for degraded features when faced with degraded images, or struggle to achieve fine-grained adaptive processing for diverse compound degradation scenarios. This results in unstable fusion performance under complex environments. To address these challenges, this paper proposes an end-to-end degradation-aware multi-modal image fusion framework (DAMFusion). The framework is built upon a two-stage autoencoder backbone, incorporating the sparse routing mechanism of a Mixture of Experts (MoE) along with heterogeneous expert structures. Guided by degradation semantics, the algorithm enables on-demand activation and task-specific optimization of expert modules, effectively balancing the trade-off between shared representation and task specificity. Enhanced fusion robustness and efficient utilization of expert modules are further achieved by introducing a mutual information based sparse constraint with KL divergence. This constraint mitigates the routing collapse problem, promotes functional diversity and load balancing among experts, and reduces redundancy and interference across expert modules, thereby strengthening the overall model stability. The proposed method demonstrates state-of-the-art (SOTA) or comparable performance on infrared-visible image fusion tasks as well as medical multi-modal image fusion tasks. In addition, cross-dataset evaluations reveal strong stability and generalization capability, highlighting its substantial practical value. The implementation is available at: DAMFusion(GitHub).

1. Introduction

Multi-modal image fusion (Karim et al., 2023) aims to effectively integrate complementary information from heterogeneous sensors or imaging modalities, producing fused images that retain the strengths of each source while offering richer representations. Such fused data provide a solid basis for scene understanding and high-level vision tasks. Among the various fusion tasks, infrared-visible image fusion (Zhang & Demiris, 2023) and medical image fusion (Hermessi et al., 2021) are the most representative. As illustrated in Fig. 1, infrared-visible image fusion leverages thermal cues from infrared images together with fine structural details from visible images, generating fused results that enhance both scene understanding and target recognition. Medical image fusion, in contrast, combines metabolic and physiological information conveyed by functional images with anatomical details revealed in structural images, yielding images that support accurate diagnosis and effective treatment. Thanks to its ability to integrate information and enhance visual quality, multi-modal image fusion has found wide applications in intelligent healthcare (Zhu et al., 2023), nighttime driving assistance (Bao et al., 2023), and object detection (Jain et al., 2023).

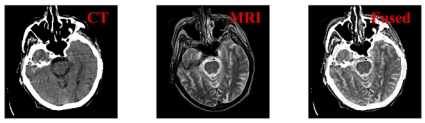
In recent years, multi-modal image fusion has garnered growing attention, with numerous algorithms proposed to pursue superior visual quality. A detailed review of representative approaches is presented in Section 2. Despite the remarkable progress, many existing approaches are designed for idealized conditions, overlooking degradations that arise in real-world acquisition. Limited capture settings often introduce noise, blur, and low resolution, substantially degrading fusion quality. For example, low illumination or overexposure distorts visible images; infrared images are susceptible to thermal noise and reduced contrast; and medical images (CT, MRI, PET) can suffer from noise, motion artifacts, or low signal-to-noise ratio. As shown in Fig. 2, directly fusing degraded inputs while ignoring these issues (Ma et al., 2020) typically yields inferior results.

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(a) Infrared-visible image fusion



(b) Medical image fusion

Fig. 1. Image fusion examples.

A straightforward remedy is a two-stage pipeline that enhances source images via preprocessing before fusion (Tang et al., 2023). While effective to a degree, this paradigm lacks awareness and adaptability: it requires manual identification of degradation types and model selection. Real images often suffer from compound degradations, such as noise, low contrast, and poor illumination. A single enhancement model cannot handle these issues, and the cooperation between multiple models is often limited. For instance, in scenes with low illumination and noise, performing brightness enhancement first amplifies noise in dark regions, whereas denoising first may blur edges or remove textures. Consequently, processing can introduce substantial distortions, making it difficult to balance the restoration of multiple degradation factors. Therefore, there is an urgent need to develop a general adaptive image fusion framework that can achieve efficient, accurate, and robust fusion under complex degradations.

To address the above challenges, this paper proposes an end-to-end degradation-aware multi-modal image fusion framework, termed DAMFusion, which integrates a Mixture of Experts (MoE) into an autoencoder (AE) backbone. Leveraging the adaptive capacity of MoE, the framework effectively combines complementary information from different modalities and addresses challenges posed by degraded images. The MoE (Mu & Lin, 2025) automatically selects optimal expert subnetworks according to input characteristics, enabling efficient multi-modal fusion while maintaining a compact architecture. In addition, degradation-aware image embeddings generated by Contrastive Language–Image Pre-training (CLIP) (Radford et al., 2021) are jointly modeled with latent features from the multi-level MoE encoder-decoder. This joint modeling allows the network to adaptively align degradation patterns with reconstruction targets in feature space, thereby achieving more fine-grained and robust fusion under semantic prior guidance. To further improve adaptability, a Kullback–Leibler (KL) divergence (Cui et al., 2024) constraint is introduced into MoE routing, which enforces sparsity and diversity in expert selection, reduces excessive overlap among inputs, and enhances robustness under compound degradations. Through this unified framework, DAMFusion is capable of effectively handling diverse degradation scenarios while preserving salient targets and capturing rich scene details, thus achieving fusion that is both robust and fine-grained.

Although this work primarily focuses on infrared-visible image fusion, the proposed approach is further extended to medical image fusion, considering that degradation issues are common across various multi-modal images. Both scenarios are included in the discussion and comparison. In summary, the main contributions of this paper are as follows:

- A novel multi-modal image fusion framework is proposed, which embeds a Mixture of Experts (MoE) into an autoencoder architecture. By exploiting the strong adaptive routing capability of MoE, the framework effectively alleviates diverse and complex degradation issues in source images.
- Two types of heterogeneous expert networks are designed: structural experts for the encoding and decoding stages, and fusion experts for the feature fusion stage. This design enables differentiated modeling and fine-grained handling of various degradation types and multi-modal inputs. In addition, a mutual information loss based on Kullback–Leibler (KL) divergence is introduced to impose sparsity on expert routing, thereby enhancing the diversity and effectiveness of expert selection.
- Extensive experimental results demonstrate that the proposed method achieves significant advantages in both infrared-visible and medical image fusion tasks, showing superior robustness and fusion performance particularly in scenarios with compound degradations.

2. Related works

This section provides a systematic review of representative image fusion methods. We organize the discussion along three lines: traditional fusion methods, deep learning-based fusion methods, and degradation-aware fusion methods, with a focus on infrared-visible and medical image fusion.

2.1. Traditional image fusion methods

Early studies on multi-modal image fusion mainly rely on multi-scale transforms, sparse representation, and saliency-driven strategies. Liu et al. (2015) propose a general framework that combines multi-scale transform with sparse representation, where the low-frequency components are fused through a sparse-coding rule and the high-frequency components are merged using a saliency-based rule. This framework subsequently becomes a widely adopted baseline for infrared-visible and medical fusion tasks. Other traditional schemes employ pyramid decomposition, wavelet analysis, or principal component analysis to extract and combine complementary information across modalities. Although such methods enjoy mathematical interpretability and require no training data, their handcrafted rules struggle to adapt when source images contain heavy noise, blur, or compound degradations frequently encountered in real-world acquisition.

2.2. Deep learning-based image fusion methods

With the rapid progress of deep learning, data-driven fusion frameworks have demonstrated superior representation capability. Autoencoder (AE)-based methods learn deep features through self-supervised reconstruction. DeepFuse (Prabhakar et al., 2017) is one of the earliest unsupervised deep models for exposure fusion with extreme exposure image pairs, training a no-reference quality metric to drive feature learning. DenseFuse (Li & Wu, 2018) introduces dense connections to preserve information across layers, while LRRNet (Li et al., 2023) leverages low-rank representation to guide effective feature fusion. Convolutional neural network (CNN) based methods, such as IFCNN (Zhang et al., 2020) and STDFusionNet (Ma et al., 2021), design task-specific architectures and fusion rules to handle different scenarios. Generative adversarial network (GAN) based methods provide an alternative perspective: FusionGAN (Ma et al., 2019) establishes an adversarial game between a generator and a discriminator to balance thermal radiation and visible texture, whereas DDcGAN (Ma et al., 2020) introduces dual discriminators to address multi-resolution fusion. Transformer-based methods, e.g., MATR (Tang et al., 2022c) and SwinFusion (Ma et al., 2022a), exploit long-range attention to capture global dependencies. In medical imaging, Zhang et al. (2022) propose an end-to-end Transformer-based framework for anatomical and functional image fusion. More recently, diffusion-based approaches such as DDFM (Zhao et al., 2023) cast image fusion as a generative modeling problem, offering improved perceptual quality.

Beyond task-specific designs, several unified frameworks attempt to handle multiple fusion tasks within a single network. U2Fusion (Xu et al., 2022) exploits unsupervised information preservation to fuse diverse modalities, and Liang et al. (2022) present a self-supervised decomposition strategy. GIFNet (Cheng et al., 2025) formulates fusion through interaction among low-level tasks within a task-agnostic

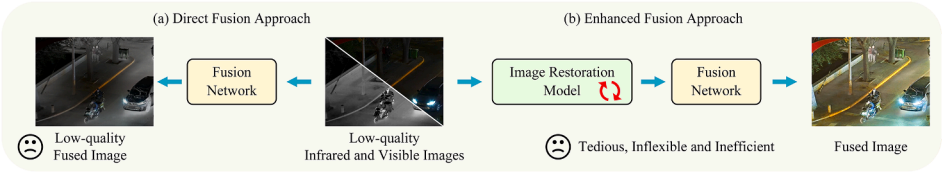


Fig. 2. Visual comparison between Direct Fusion and Enhanced Fusion approach.

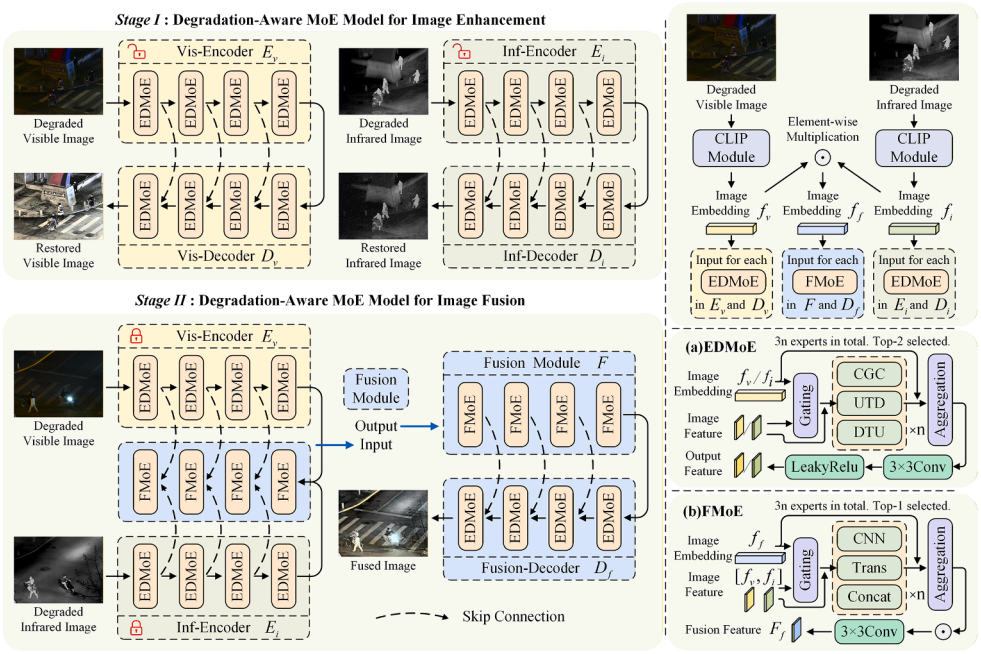


Fig. 3. Overall architecture of the proposed DAMFusion framework. Stage I trains the modality-specific encoders (E_v, E_i) and decoders (D_v, D_i) on image restoration; Stage II freezes the encoders and trains the fusion module together with the decoder D_f for end-to-end fusion. Degradation-aware embeddings produced by CLIP guide both the encoder-decoder branch (via EDMoE) and the fusion branch (via FMoE), enabling adaptive expert routing under diverse degradation conditions.

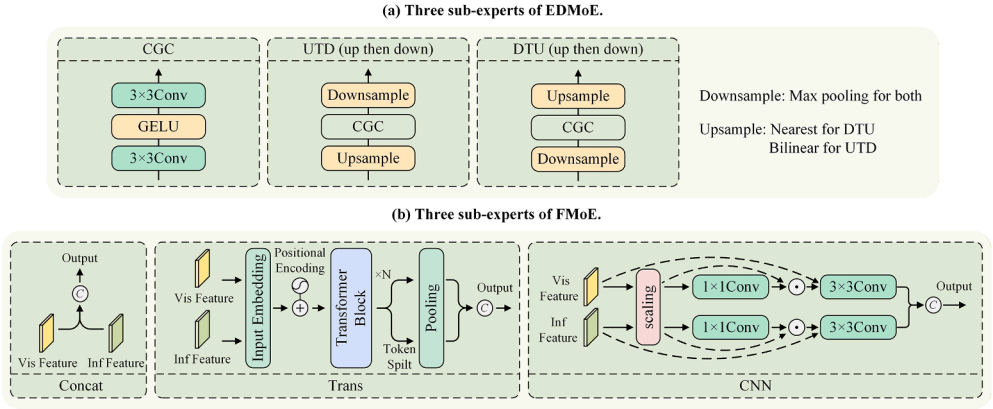


Fig. 4. Expert structures for EDMoE and FMoE.

framework. SSPFusion (Liu et al., 2024) further introduces a semantic structure-preserving formulation that retains modality-specific structures during multi-modality fusion. Although these methods achieve impressive performance on clean inputs, they are typically optimized under idealized acquisition conditions and lack explicit mechanisms for handling diverse degradations.

2.3. Degradation-aware image fusion methods

To improve robustness in real-world scenes, recent efforts have begun to explicitly handle image degradations during fusion. Bhutto et al. (2022) combine adaptive contrast enhancement with a transfer-learning-based deep network for infrared-visible image fusion. DIVFusion (Tang et al., 2023) integrates low-light enhancement with infrared-visible fusion to recover visibility in dark environments. Wang et al. (2023) propose an enlighten multi-scale network tailored for dark scenes. DAAF (Zhang et al., 2025) presents a degradation-aware adaptive fusion framework with local-global modeling, but its performance heavily depends on accurate degradation feature extraction and is hard to generalize under compound degradations. Text-IF (Yi et al., 2024) leverages semantic text guidance to drive degradation handling, yet it relies on external textual descriptions and supports primarily single-

type degradations such as low light, blur, or low contrast. DAFusion (Wang et al., 2025) introduces implicit degradation estimation for adaptive guided fusion, though fine-grained adaptation under compound degradations remains challenging. Yi et al. (2024) further propose an instruction-driven fusion paradigm that tailors infrared-visible fusion to diverse downstream tasks, improving task-level adaptivity.

Despite these advances, most existing methods either focus on a specific degradation type or assume a single, easily separable degradation pattern, making them less effective under compound degradations frequently encountered in practice. Motivated by these limitations, the proposed DAMFusion framework introduces a unified architecture with degradation-aware semantic priors and a sparse Mixture of Experts routing mechanism, enabling fine-grained adaptive fusion under diverse and compound degradation conditions.

3. Method

As shown in Fig. 3, the proposed model is built upon an autoencoder backbone with a two-stage learning strategy. *Stage I* emphasizes degradation modeling, where image restoration is adopted as a proxy task. Degraded images are fed into the autoencoder, allowing the encoder to explicitly capture various degradation patterns during the restoration process and yielding modality-adaptive encoders E_v and E_i. *Stage II* addresses fusion and reconstruction, where degraded images are processed by the frozen encoders E_v and E_i to extract multi-scale features with degradation compensation. The fusion module performs adaptive processing on aligned branches to obtain fused features, which are progressively passed to the decoder D_f through skip connections for high-quality image reconstruction. Inference requires only *Stage II*, enabling efficient end-to-end fusion within a single forward pass.

To enhance coverage of diverse degradations, a pretrained CLIP module (Radford et al., 2021) provides degradation-aware image embeddings as semantic priors. These embeddings condition both feature modeling and fusion, enabling reliable identification of individual degradations and adaptive category selection in compound cases. The mechanism aligns naturally with the dynamic expert routing of MoE, supporting fine-grained perception and modeling of degradation.

The framework further introduces two heterogeneous sparse expert networks: Encoder-Decoder MoE (EDMoE) and Fusion MoE (FMoE). Structural and implementation details are provided in Section 3.1. In addition, a KL divergence based mutual information loss is imposed on expert routing to enforce controllable sparsity, improve diversity, and enhance robustness, as elaborated in Sections 3.2 and 3.3.

3.1. EDMoE and FMoE

As shown in Fig. 3, two types of Mixture of Experts (MoE) networks are designed: Encoder-Decoder MoE (EDMoE) for encoding/decoding and Fusion MoE (FMoE) for multi-scale fusion. EDMoE forms the core of the encoders (E_v, E_i) and decoders (D_v, D_i, D_f), where adaptive routing combines expert outputs to ensure robust feature extraction and image reconstruction. FMoE, as the main unit of the fusion module, facilitates deep cross-modal interaction and fusion. Leveraging multi-scale modeling and sparse routing, it fully exploits modality complementarity and enhances both detail fidelity and global consistency of the fused images. Both EDMoE and FMoE comprise three types of sub-expert networks. Multiple instances are deployed in parallel, and their outputs are aggregated by routing weights. Although structurally identical, the sub-experts exhibit functional diversity owing to independent parameter training. The structure is shown in Fig. 4.

(1) **EDMoE** consists of three types of expert networks: CGC, UTD, and DTU. The CGC expert acts as a relatively simple structural unit that reliably performs basic feature extraction, ensuring the model's robust representation capability under various input conditions. The UTD expert is more suitable for processing regions with concentrated local feature distributions in degraded visible-light and infrared images. It enhances local areas through upsampling, followed by feature extraction and downsampling, which aids in capturing fine textures and local details, thus compensating for the loss of critical information in low-resolution or blurry scenes. In contrast, the DTU expert is designed for more complex structures in infrared and visible-light images. It first downscales to aggregate global contextual information, then performs feature extraction and upsampling for reconstruction. This process better models cross-scale dependencies, enhancing the model's global perception and fusion in complex scenarios. Through the complementary design of these three experts, the model can adapt to various degradation types for feature extraction and reconstruction, accurately capturing both local details and global semantic and structural information, thus ensuring comprehensive and robust modeling capabilities.

(2) **FMoE** consists of three types of expert networks: CNN, Trans, and Concat. The CNN expert uses a lightweight convolutional structure, ideal for regions with large pixel distributions and smooth overall fusion effects. It reliably extracts key features while maintaining robustness. The Trans expert, based on the attention mechanism, tokenizes and maps two input features before sending them to a self-attention module. This mechanism enables long-range dependency modeling between cross-modal features, particularly for precise fusion of local details and textures in degraded infrared and visible-light images. In contrast, the Concat expert performs simple channel concatenation, effectively processing regions with minimal differences between infrared and visible light, especially in medical image tasks where large areas of low information (e.g., black backgrounds) are present. This approach completes stable fusion without complex modeling.

3.2. Gating network

In a Mixture of Experts (MoE) network, the core idea is to select and weigh multiple experts using a gating mechanism. In its general form, for an input x, the output of the MoE layer can be expressed as:

$$y = \sum_{i=1}^{M} w_i f_i(x) \tag{1}$$

where M is the total number of experts, $f_i(\cdot)$ represents the ith expert sub-network, and w_i is the corresponding gating weight. Unlike traditional dense weighting, this paper introduces a sparse gating mechanism in which only a small number of experts are activated for computation while the others remain silent. This significantly reduces computational overhead while maintaining expressive capability.

The routing process is designed to depend not only on the low-level features of the input image but also on the degradation-related semantic information. To achieve this, the gating network incorporates Image Embedding of degraded images: f_v, f_i, f_f. Specifically, after pooling, the input features are concatenated with the semantic embeddings and then passed through a fusion mapping, resulting in the fused feature representation z. On this basis, the response strength of each expert is computed as:

$$l = zW_g \tag{2}$$

where W_g is a learnable reinforcement matrix. The next step involves normalizing l to obtain the probability distribution for each expert. Directly using l for routing may lead to certain experts dominating for extended periods, especially in the early stages of training. To address this issue, noise perturbation is introduced on the logits to enhance exploration. Injecting Gaussian noise into the logits is a standard practice in MoE routing (Mu & Lin, 2025), and it encourages expert exploration during early training:

$$l = l + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{3}$$

where σ is the noise magnitude calculated as:

$$\sigma = \text{softplus}(zW_n) + \delta, \quad \delta = 10^{-6} \tag{4}$$

Here, σ controls the noise intensity, and δ is a numerical stability term. The introduction of noise improves the model's exploration capabilities, preventing experts from prematurely converging to a single mode and promoting diversity and balance in expert utilization.

For sparse computation, the paper employs a Top-K strategy in the gating network. Only the top K experts with the highest probabilities are retained in the set S, with their probabilities re-normalized. The weights of all other experts are set to zero. In our implementation, we set $K = 2$ for EDMoE and $K = 1$ for FMoE, which balances representation diversity and routing sparsity according to the role of each module. The weight w_i in Eq. (1) is expressed as:

$$w_i = \begin{cases} \dfrac{e^{l_i}}{\sum_{j\in S} e^{l_j}}, & i \in S \\ 0, & i \notin S \end{cases} \tag{5}$$

This design reduces computational complexity during the forward pass while enhancing the discriminability of the routing process. It encourages complementary roles among experts, rather than distributing the workload evenly.

The Top-K strategy alone may lead to "routing collapse," where certain experts are selected frequently while others remain idle. To resolve this, an inter-information constraint based on KL divergence is introduced in each MoE gating unit (including EDMoE and FMoE). This constraint directly influences the gating probabilities, controlling sparsity across different modalities and degradation types, and improving balance and diversity in expert utilization. The basic form of this constraint is:

$$\mathrm{KL}\big(P(E|T) \,\|\, P(E)\big) = \sum_{i=1}^{M} P(E=i|T) \log \frac{P(E=i|T)}{P(E=i)} \tag{6}$$

where $P(E)$ is the marginal distribution of experts within the batch, and $P(E|T)$ is the expert distribution conditioned on a given modality or degradation type T. The mutual information is further defined as:

$$\tilde{MI} = \sum_{T} P(T)\,\mathrm{KL}\big(P(E|T) \,\|\, P(E)\big) \tag{7}$$

By explicitly incorporating this mutual information constraint into the routing process, the model achieves a sparse yet stable expert allocation across multiple modalities and degradation conditions. The forward pass follows the Top-K strategy, and the mutual information term only regularizes the gating probabilities during training, with no impact on the inference process.

3.3. Loss

In terms of loss function design, a two-stage training strategy is adopted. The first stage primarily trains the two modality encoders E_v and E_i to ensure effective recovery and representation of degraded source images in a multi-scale feature space, laying a stable foundation for subsequent fusion. The second stage freezes the parameters of E_v and E_i, and trains the fusion branch and its decoder, enabling the model to achieve information complementarity and detail enhancement under multi-modal conditions. The total loss functions for the two stages are defined as:

$$\begin{cases} L_I = \lambda_1 L_{\mathrm{MI}} + \lambda_2 L_{\mathrm{ssim}} + \lambda_3 L_{\mathrm{pix}}, \\ L_{II} = \lambda_4 L_{\mathrm{MI}} + \lambda_5 L_{\mathrm{ssim}} + \lambda_6 L_{\mathrm{pix}} + \lambda_7 L_{\mathrm{grad}}. \end{cases} \tag{8}$$

where $\{\lambda_1, \lambda_2, \ldots, \lambda_7\}$ are the weights of each loss component.

The mutual information loss L_{MI} primarily affects the gating mechanism of the Mixture of Experts network, aiming to improve the sparsity and diversity of routing. By introducing KL divergence constraints under different modalities or degradation conditions, L_{MI} encourages a more dispersed gating probability distribution, effectively utilizing more experts. Based on Eq. (7), the calculation is as follows:

$$L_{\mathrm{MI}} = -\lambda_{\mathrm{MI}} \cdot \tilde{MI} \tag{9}$$

where $\tilde{MI}$ denotes the mutual information value estimated by KL divergence, and λ_{MI} is the sparsity weight. In this model, each MoE gating unit introduces a mutual information loss, so the total mutual information loss is the sum of all gating network loss terms.

The pixel loss L_{pix} constrains the difference in pixel intensity between the model's output and the target image. By minimizing the pixel-wise difference between the predicted result and the reference image, the model effectively preserves the consistency of brightness and detail information. In this study, pixel loss is defined using the L_1-norm:

$$L_{\mathrm{pix}} = \|\hat{I} - I\|_1 \tag{10}$$

where $\hat{I}$ is the model output and I is the reference target image.

The structural similarity loss L_{ssim} measures the similarity in structural information between the predicted and reference images. It considers brightness, contrast, and structure to reflect the perceptual quality differences between the two images. In this study, structural similarity loss is defined as:

$$L_{\mathrm{ssim}} = 1 - \frac{(2\mu_{\hat{I}}\mu_I + C_1)(2\sigma_{\hat{I}I} + C_2)}{(\mu_{\hat{I}}^2 + \mu_I^2 + C_1)(\sigma_{\hat{I}}^2 + \sigma_I^2 + C_2)} \tag{11}$$

where $\mu_{\hat{I}}$ and μ_I represent the means of images $\hat{I}$ and I, $\sigma_{\hat{I}}^2$ and σ_I^2 represent their variances, $\sigma_{\hat{I}I}$ is the covariance, and C_1 and C_2 are constants for numerical stability. In Stage I, the pixel loss L_{pix} and structural similarity loss L_{ssim} measure the difference between the source and recovered images, while in Stage II, they calculate the difference between the fused image and the two source images.

The gradient loss L_{grad} constrains the similarity in edges and details between the predicted and reference images. It emphasizes the structural contours and texture boundaries of the images, helping prevent blurring or loss of details in the fusion result. For the two modalities, the gradient with the stronger response at each point is selected as the reference gradient for the fused image. In this study, the gradient loss is defined using the L_1-norm as:

$$\begin{cases} L_{\mathrm{grad}} &= \|\nabla\hat{I} - \nabla I\|_1, \\ \nabla I &= \max(\nabla I_1, \nabla I_2). \end{cases} \tag{12}$$

where ∇ represents the gradient operator used to extract the edges and structural features of the image, and $\hat{I}$, I_1, and I_2 represent the fused image and the two source images, respectively.

4. Experiments

Experiments are crucial for testing the algorithm's performance and feasibility. This chapter systematically evaluates the proposed algorithm using standard datasets and metrics to validate its effectiveness and robustness.

4.1. Implementation detail

(1) **Experimental Configuration**: All experiments in this study were conducted in a unified hardware environment, ensuring that all comparison methods were evaluated under the same conditions. The specific experimental configuration is shown in Table 1.

(2) **Evaluation metrics**. This research employs nine widely used metrics from multiple perspectives to quantitatively assess the performance of fusion algorithms. These metrics fall into three categories: information theory-based, image feature-based, and human visual perception-based, as summarized in Table 2. For all nine metrics, higher values indicate better fusion quality.

(3) **Training details**. This research adopts a two-stage training pipeline. In the first stage, the encoder is trained with diverse augmentations using a batch size of 1, a learning rate of 1×10^{-4}, and 1×10^3 epochs, where the input images are randomly cropped to 64×64. In the second stage, the encoder from Stage I is frozen, and the fusion network is trained end-to-end under the same hyperparameters with random 128×128 crops. For infrared-visible image fusion, 300 paired images are uniformly sampled from four public visible-infrared datasets,

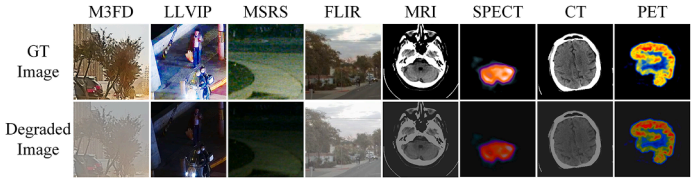


Fig. 5. Example of the image restoration dataset for *Stage I*. *Stage II* directly uses the original dataset for fusion training.

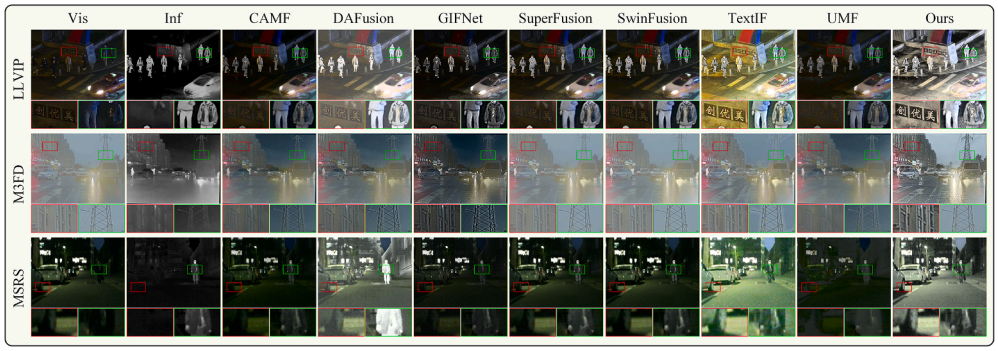


Fig. 6. Visual comparison of Direct Fusion on the infrared-visible dataset.

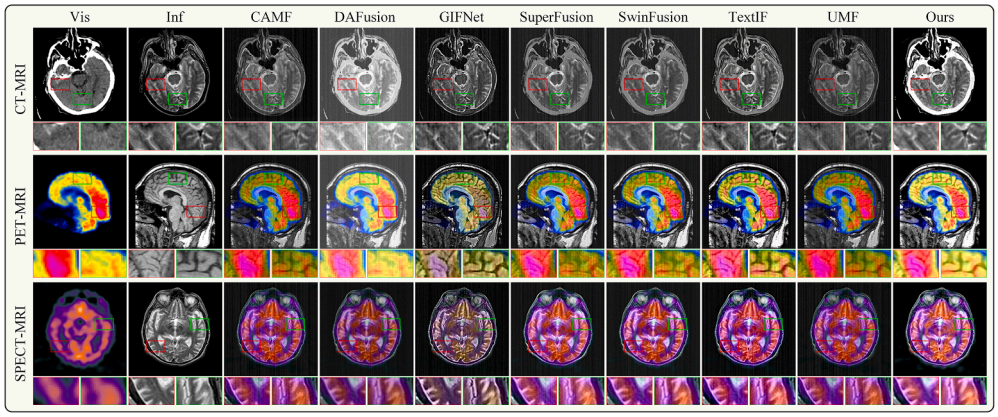


Fig. 7. Visual comparison of Direct Fusion on the medical dataset.

Table 1
Experimental configuration.

Datasets	1.LLVIP (Jia et al., 2021) 2.M3FD (Liu et al., 2022) 3.MSRS (Tang et al., 2022b) 4.FLIR (FLIR Systems, 2019) 5. Harvard Medical Dataset (Vidoni, 2012) { CT-MRI PET-MRI SPECT-MRI	
Hardware	NVIDIA GeForce RTX 3090 GPU	
Software	Python 3.10 Cuda 11.3 Pytorch 1.11.0	OpenCV 4.5.5.64 Numpy 1.26.1

while for medical image fusion, 700 paired images are randomly selected from three categories of the Harvard Medical Dataset.

Notably, *Stage I* training with image augmentations requires ground truth (GT). For visible images, dataset-specific strategies are employed: LLVIP, dominated by low-light conditions, uses original images as degraded inputs, with low-light compensation results (Tang et al., 2022b) serving as GT; M3FD, primarily affected by low contrast and blur, uses contrast-enhanced images as GT, while degraded inputs are simulated through brightness attenuation, blurring, and additive noise; MSRS, exhibiting both low-light and low-contrast issues, generates pseudo-GT by applying contrast adjustment to low-light compensated images (Tang et al., 2022b); FLIR, characterized by low contrast and noise, creates GT via brightness enhancement and contrast adjustment. In contrast, infrared images typically experience milder degradation, so the original images are used as GT, and degraded versions are synthesized by adding mild blur and noise. For medical images, the process is simpler: the Harvard Medical Dataset, being of high quality, uses the original images as GT, with degraded samples generated by applying contrast and brightness adjustments, blurring, and additive noise. Representative training samples are shown in Fig. 5, with a portion of the dataset reserved for validation.

(4) **Comparison methods**. This research compares the proposed approach with seven recent state-of-the-art (SOTA) methods to comprehensively evaluate its effectiveness, including CAMF (Tang et al., 2024), DAFusion (Wang et al., 2025), GIFNet (Cheng et al., 2025), SuperFusion (Tang et al., 2022a), SwinFusion (Ma et al., 2022a), Text-IF (Yi et al., 2024), and UMF (Wang et al., 2022). These methods represent the frontier of image fusion, and the comparison highlights both the effectiveness of the proposed method and its advancement in degraded image

Table 2
Evaluation metrics.

Category	Metrics	Attention information
Information Theory-Based	Entropy (EN ↑) (Sheikh & Bovik, 2006)	Image complexity and information content
	Mutual Information (MI ↑) (Pluim et al., 2003)	Shared information between images
Image Feature-Based	Average Gradient (AG ↑) (Ma et al., 2016)	Image sharpness and detail preservation
	Spatial Frequency (SF ↑) (Eskicioglu & Fisher, 1995)	Image texture and detail level
	Standard Deviation (SD ↑) (Qu et al., 2002)	Contrast and intensity variations across the images
	Structural Content Difference (SCD ↑) (Aslantas & Bendes, 2015)	Structural similarity between source and fused images
Human Visual Perception-Based	Correlation Coefficient (CC ↑) (Han et al., 2013)	Similarity and consistency between source and fused images
	Visual Information Fidelity (VIF ↑) (Sheikh & Bovik, 2006)	Preservation of visual information aligned with human perception
	Absolute Error Fusion Quality (Qabf ↑) (Piella & Heijmans, 2003)	Fusion quality based on absolute error

Table 3
Quantitative analysis of Direct Fusion. The best and second results are highlighted in red and blue, respectively.

	LLVIP									*CT-MRI*								
	SD	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*	*SD*	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*
CAMF	35.94	2.53	0.55	3.17	**0.76**	1.37	6.84	0.28	10.16	54.03	2.45	0.31	7.04	0.80	0.86	6.76	0.29	19.66
GIFNet	42.25	2.45	0.57	6.31	0.73	1.43	6.81	0.45	25.37	52.72	2.36	0.30	**9.90**	0.76	0.60	6.85	0.37	**30.44**
SuperFusion	45.73	3.06	0.80	5.10	0.72	1.49	7.16	**0.60**	17.09	48.92	2.56	0.32	7.36	0.78	0.51	6.84	0.29	21.42
SwinFusion	46.94	**3.94**	0.89	5.97	0.72	1.28	7.14	**0.67**	20.81	48.00	2.63	0.34	7.21	0.78	0.48	6.82	0.29	21.03
UMF	30.97	2.55	0.51	3.77	0.73	1.01	6.57	0.36	15.32	43.22	2.48	0.33	7.06	0.77	0.21	6.80	0.32	20.36
DAFusion	50.92	3.11	0.83	5.35	**0.74**	**1.68**	7.32	0.52	18.45	**58.01**	**2.65**	0.35	7.39	**0.81**	**0.94**	**6.91**	0.33	22.62
Text-IF	**52.45**	3.10	**0.92**	**9.24**	0.63	1.10	**7.43**	0.34	**30.00**	54.64	2.45	**0.39**	**9.67**	0.76	0.67	**7.02**	**0.43**	28.33
Ours	**65.11**	**3.86**	**0.89**	**11.15**	0.64	**1.52**	**7.87**	0.35	**36.10**	**86.92**	**2.80**	**0.45**	7.55	**0.85**	**1.62**	4.60	**0.50**	**31.50**
	M3FD									*PET-MRI*								
	SD	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*	*SD*	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*
CAMF	32.79	3.01	0.65	2.43	0.54	**1.71**	6.89	0.45	7.91	72.58	2.40	0.44	9.29	**0.84**	**0.99**	7.07	0.46	25.10
GIFNet	**36.45**	1.91	0.81	**7.12**	0.55	1.48	**6.90**	0.26	**15.91**	67.70	2.18	0.36	10.77	0.81	0.62	6.97	0.49	**37.91**
SuperFusion	16.59	**3.74**	0.83	2.32	0.49	1.03	5.68	0.52	8.27	69.64	2.61	0.43	10.26	0.76	0.23	**7.16**	0.52	27.47
SwinFusion	18.92	3.34	**0.87**	2.48	0.55	1.28	5.84	**0.54**	8.76	70.42	2.70	0.50	10.93	0.78	0.32	**7.14**	0.61	29.26
UMF	28.00	3.41	0.67	1.94	0.51	1.46	6.61	0.42	6.56	57.85	2.53	0.41	9.50	0.78	0.05	7.11	0.48	29.18
DAFusion	30.01	2.64	0.82	3.64	**0.57**	1.65	6.80	0.53	11.70	55.43	2.51	0.50	7.93	0.82	0.20	6.96	0.40	22.39
Text-IF	24.14	2.36	**0.85**	3.76	0.55	1.31	6.35	**0.61**	12.69	**77.34**	**2.63**	**0.53**	**13.12**	0.77	0.63	7.11	**0.65**	**36.95**
Ours	**51.15**	**3.45**	0.84	**7.17**	**0.56**	**1.66**	**7.32**	0.27	**17.86**	**89.41**	**2.89**	**0.56**	**12.68**	**0.82**	**1.48**	5.61	**0.63**	31.97
	MSRS									*SPECT-MRI*								
	SD	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*	*SD*	*MI*	*VIF*	*AG*	*CC*	*SCD*	*EN*	*Qabf*	*SF*
CAMF	18.45	1.76	0.55	1.24	**0.72**	1.27	5.00	0.26	4.63	54.30	2.54	0.40	6.55	0.88	0.92	6.60	0.42	18.16
GIFNet	22.36	1.64	0.59	1.67	0.67	1.32	4.81	0.36	8.44	46.00	2.31	0.36	**8.45**	0.83	0.08	6.46	0.47	**26.01**
SuperFusion	29.15	3.09	0.97	1.90	0.69	1.78	5.55	**0.58**	7.30	50.62	2.63	0.42	7.24	0.83	0.12	6.68	0.48	20.29
SwinFusion	29.69	3.25	1.00	2.02	0.68	**1.78**	5.60	**0.58**	7.46	48.30	**2.72**	0.46	7.35	0.82	0.10	**6.71**	0.53	20.24
UMF	13.85	1.73	0.32	1.33	0.53	0.54	4.89	0.23	4.86	35.80	2.33	0.35	6.21	0.83	0.41	6.54	0.39	18.28
DAFusion	39.40	2.11	1.21	**4.22**	0.64	1.51	6.62	0.25	**11.38**	**57.27**	2.61	0.43	6.73	**0.90**	**1.25**	**6.85**	0.43	19.36
Text-IF	**43.99**	**3.29**	**1.25**	3.48	0.50	0.90	**7.08**	0.29	9.32	53.09	2.59	**0.50**	**8.48**	0.83	0.24	6.50	**0.59**	24.28
Ours	**49.78**	**3.32**	**1.25**	**3.98**	**0.69**	**1.63**	**7.17**	0.24	**11.35**	**62.85**	**2.97**	**0.56**	6.55	**0.89**	**1.43**	5.11	**0.59**	**24.89**

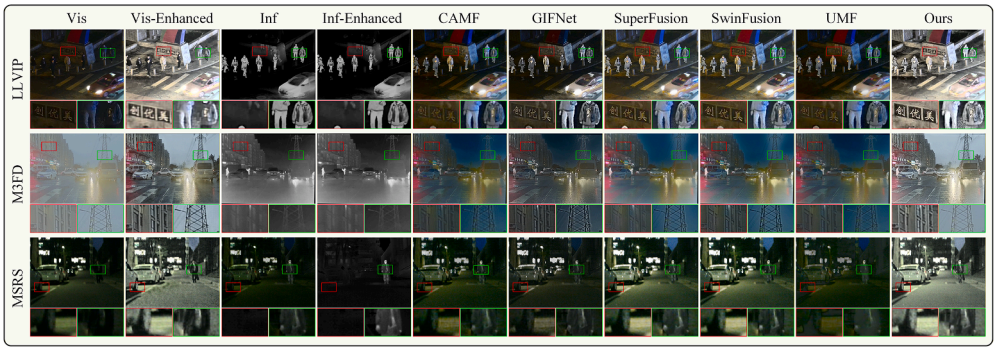


Fig. 8. Visual comparison of Enhanced Fusion on the infrared-visible dataset.

fusion. It should be noted that, except for DAFusion and Text-IF, the other SOTA methods do not explicitly consider image degradation.

4.2. Experimental results

The experimental design includes two fusion strategies, Direct Fusion and Enhanced Fusion, to thoroughly validate the effectiveness and robustness of the proposed method. In the Direct Fusion experiment, source images with degradation issues are directly fused. In the Enhanced Fusion experiment, the source images are first restored to address degradation before fusion.

(1) **Direct Fusion**. Direct fusion is a common approach used by most methods, where degradation is not explicitly modeled or preprocessed. The experimental data is presented in Table 3.

The experiments show that fusion methods that explicitly address degradation outperform those that do not. Among the methods that handle degradation, the proposed method demonstrates the most robust and overall optimal performance, achieving the best results across multiple evaluation metrics. This fully demonstrates its advantages in terms of robustness and generalization. The method effectively handles fusion tasks involving degraded images, generating high-quality fusion results and showing significant superiority in the field of degraded image fusion. It is worth noting, however, that the Qabf metric on the infrared-visible datasets does not always reach the best score: methods that aggressively sharpen edges (e.g., SwinFusion, SuperFusion) can attain higher Qabf since the metric is dominated by edge-preservation strength. Our method instead balances edge preservation with structural fidelity and contrast, which leads to slightly lower Qabf but consistent advantages across the remaining eight metrics. In medical image fusion tasks, the proposed method also performs outstandingly. In particular, it achieves the highest values in four core metrics: SD, MI, VIF, and SCD, indicating that the method not only preserves structural integrity but also effectively enhances image quality and detail representation. Its high precision and accurate detail recovery capabilities further highlight its immense potential in medical image processing.

This study compared the visual results of various algorithms to validate the performance of the proposed method under different degradation scenarios. As shown in Fig. 6, the proposed method performs exceptionally well in dehazing and low-light enhancement tasks. Compared to other methods, it restores image details more effectively under low light and complex degradation conditions, showing significant advantages in structural fidelity and visual perception. Overall, the proposed method not only maintains good image structural consistency but also enhances details more accurately, achieving better visual effects. The results in Fig. 7 further demonstrate the excellent performance of the proposed method in multi-modal fusion of degraded medical images. Especially on the CT-MRI, PET-MRI, and SPECT-MRI multi-modal datasets, the proposed method fully integrates the complementary information from different modalities, effectively preserving key structural features and details. Compared to other methods, the proposed method shows more significant advantages in improving image clarity and restoring details, particularly in complex medical imaging scenarios (especially CT and MRI), fully proving its wide applicability and potential in multi-modal medical image fusion.

(2) **Enhanced Fusion**. Preprocessing source-image degradations before fusion is an effective way to mitigate adverse effects and improve fused-image quality. To highlight the proposed algorithm's advantages, this study preprocesses competing methods that do not explicitly model degradation, resolves their degradations, and then compares them with the proposed method. Text-IF and DAFusion already incorporate degradation modeling and handling and are therefore excluded. For infrared images, manually added noise is removed. For the low-light datasets LLVIP and MSRS, SCI (Ma et al., 2022b) is applied to brighten the visible images; for M3FD, dehazing is performed with DCP (He et al., 2010). For the medical datasets, clean, noise-free, blur-free, and normally exposed source images are directly used as the Enhanced Images for comparison. Results are reported in Table 4.

Experimental results show that after enhancing the source images, the performance of all comparison algorithms improves. However, the proposed method still maintains comprehensive optimal performance across different datasets and multiple metrics, fully verifying its adaptability and robustness in various degradation scenarios. Although the two-stage processing strategy can improve the performance of other methods to some extent, the proposed algorithm still demonstrates superior fusion performance, highlighting its advantage in multi-modal degraded image fusion tasks.

The visual results further support this. In Fig. 8, the proposed method effectively enhances the details and contrast in the fusion of degraded infrared-visible images, significantly improving distortions caused by low light and blur, resulting in clearer and more detailed outputs. In Fig. 9, during medical image fusion, the proposed method not only preserves structural integrity but also significantly enhances detail recovery and image visibility, effectively denoising and improving quality even under low contrast and blurred conditions.

Overall, the key to degraded image fusion lies in addressing degradation issues. The multi-task adaptive strategy proposed in this study demonstrates excellent flexibility and robustness in various degradation scenarios, providing strong support for improving fusion quality.

4.3. Extension experiment

Image fusion is often used as an upstream task in many computer vision applications, such as object detection (Jain et al., 2023), image segmentation, and image classification. The effective information enhancement and aggregation provided by image fusion not only contribute to visual perception but also improve the performance of machine vision. Beyond perceptual and machine-vision quality, the practical deployment of fusion methods also depends on computational cost.

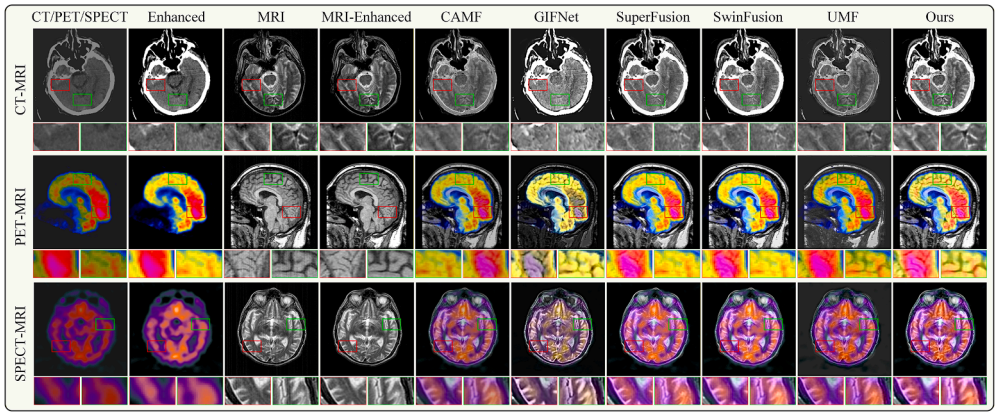


Fig. 9. Visual comparison of Enhanced Fusion on the medical dataset.

Table 4
Quantitative analysis of Enhanced Fusion. The best and second results are highlighted in red and blue, respectively.

	LLVIP									*CT-MRI*								
	SD	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***	***SD***	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***
CAMF	43.17	2.38	0.58	4.40	**0.76**	**1.61**	7.17	0.36	13.47	63.36	2.64	0.33	5.42	**0.82**	1.17	**5.20**	0.32	19.50
GIFNet	**54.98**	2.45	0.63	**8.74**	0.73	**1.66**	7.28	0.41	**33.45**	82.27	2.90	0.36	**8.42**	0.75	1.25	5.08	0.36	**40.25**
SuperFusion	52.22	**3.87**	**0.90**	7.13	0.70	1.48	7.37	**0.52**	23.15	85.04	**3.16**	**0.51**	6.73	0.79	1.51	4.83	**0.51**	28.24
SwinFusion	53.62	**3.98**	**0.98**	8.02	0.70	1.49	**7.38**	0.48	27.14	**85.79**	**3.22**	**0.53**	6.88	0.79	**1.58**	4.93	**0.52**	28.96
UMF	40.48	2.43	0.66	5.59	**0.75**	1.50	7.07	**0.52**	22.07	50.31	2.69	0.35	6.69	0.79	0.54	**5.55**	0.43	28.89
Ours	**65.11**	3.86	0.89	**11.15**	0.64	1.52	**7.87**	0.35	**36.10**	**86.92**	2.80	0.45	**7.55**	**0.85**	**1.62**	4.60	0.50	**31.50**
	M3FD									***PET-MRI***								
	SD	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***	***SD***	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***
CAMF	28.58	2.02	0.62	3.14	**0.57**	**1.55**	6.70	0.34	10.06	76.01	2.69	0.48	6.85	**0.86**	1.32	**6.00**	0.36	19.95
GIFNet	36.45	1.91	0.81	**7.12**	0.55	1.48	6.90	0.26	15.91	80.13	2.66	0.41	8.03	**0.84**	1.34	5.57	0.33	26.49
SuperFusion	33.96	**3.89**	**1.00**	4.54	0.53	1.35	6.93	**0.41**	15.50	83.71	2.83	0.53	9.18	0.81	1.23	5.87	0.56	26.87
SwinFusion	**36.83**	2.52	**1.23**	5.77	0.53	1.37	**7.03**	0.34	**19.20**	**88.86**	**3.04**	**0.63**	**10.13**	0.83	**1.51**	5.95	**0.66**	**30.18**
UMF	27.46	2.28	0.85	3.56	0.56	1.52	6.60	**0.47**	13.36	60.19	2.48	0.42	7.85	0.82	0.41	**6.24**	0.43	26.13
Ours	**51.15**	**3.45**	0.84	**7.17**	**0.56**	**1.66**	**7.32**	0.27	**17.86**	**89.41**	**2.89**	**0.56**	**12.68**	0.82	**1.48**	5.61	**0.63**	**31.97**
	MSRS									***SPECT-MRI***								
	SD	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***	***SD***	***MI***	***VIF***	***AG***	***CC***	***SCD***	***EN***	***Qabf***	***SF***
CAMF	23.43	2.04	0.70	1.74	**0.69**	1.47	5.60	0.35	5.70	60.34	2.76	0.43	4.49	0.91	**1.50**	**5.49**	0.39	13.92
GIFNet	29.98	1.82	0.74	2.41	0.66	**1.62**	5.53	**0.38**	**10.94**	56.42	2.77	0.40	**6.63**	0.88	1.06	5.06	0.46	**23.27**
SuperFusion	41.51	**3.87**	1.24	2.77	0.61	1.32	6.44	**0.43**	9.43	59.42	**3.14**	**0.51**	5.50	0.88	1.21	5.21	0.54	17.81
SwinFusion	**41.64**	**4.13**	**1.30**	**2.91**	0.60	1.28	**6.50**	0.40	9.56	**60.38**	**3.24**	0.46	5.55	**0.88**	1.33	5.34	**0.58**	17.43
UMF	17.47	1.66	0.51	1.73	0.65	1.03	5.25	0.34	6.00	37.50	2.67	0.39	4.71	0.85	0.22	**5.63**	0.45	15.44
Ours	**49.78**	3.32	**1.25**	**3.98**	**0.69**	**1.63**	**7.17**	0.24	**11.35**	**62.85**	2.97	**0.56**	**6.55**	**0.89**	**1.43**	5.11	**0.59**	**24.89**

This subsection therefore evaluates the proposed framework from two complementary perspectives: (i) downstream object detection performance, and (ii) computational efficiency in terms of model size, FLOPs, and inference time.

4.3.1. Downstream object detection

This study evaluates the quality of fused images generated by the proposed algorithm through the object detection task on the M3FD and LLVIP datasets. To ensure a comprehensive comparison and analysis, enhanced fused images are used for the experiments and discussions, employing four widely used evaluation metrics (Jain et al., 2023): Precision, Recall, mAP50, and mAP50:95. Under the same experimental configuration as shown in Table 1, this study uses a pre-trained YOLOv8 model (Jocher et al., 2023) to validate and assess the fused images generated by each comparison algorithm. The visual comparisons of the algorithms are shown in Fig. 10.

The proposed method exhibits strong robustness across multi-scene object detection tasks, with virtually no false or missed detections, while maintaining high confidence scores and stable predictions. This further indicates that the fused images generated by our approach possess high information integrity and discriminability, effectively supporting the reliability of downstream vision tasks. Furthermore, after normalizing the individual metrics, we plotted a radar chart; the results are shown in Fig. 11. It shows that our method delivers stable performance across four detection metrics, yielding reliable detection quality on fused images. Note that the LLVIP dataset contains only a single class (person).

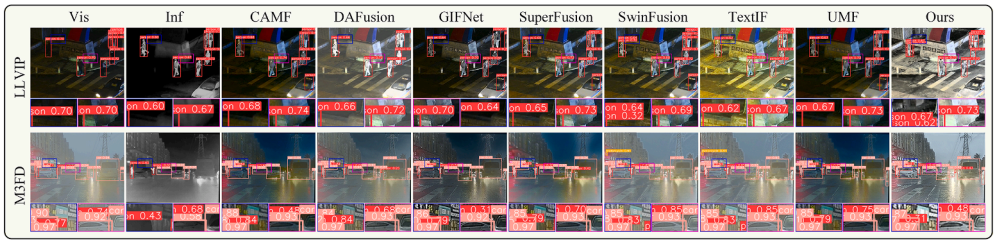


Fig. 10. Visual comparison of object detection. The LLVIP dataset only provides labels for the "Person" class.

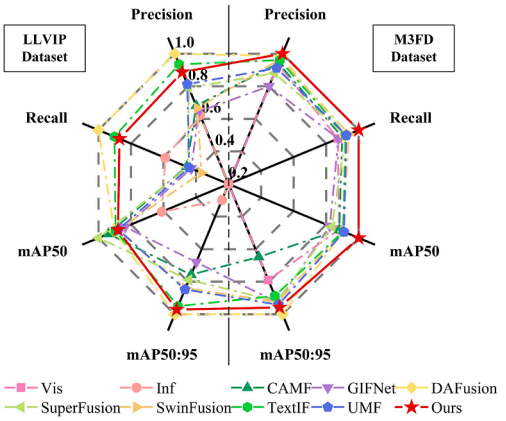


Fig. 11. Radar chart of normalized detection metrics. The LLVIP dataset only provides labels for the "Person" class.

The leading scores of DAFusion and Text-IF on LLVIP mainly arise from retaining more infrared cues. In particular, brightness boosts enhance target saliency and make objects easier for the detector to capture. However, in multi-class or fine-grained scenarios, this strategy can amplify background clutter and weaken structural textures, limiting its advantage. In contrast, our approach improves detectability while better preserving structural fidelity and visual naturalness, achieving a stronger balance between detection accuracy and perceptual quality. Moreover, even against methods that first restore the source images before fusion (a more favorable setting), our approach attains superior overall performance, indicating more discriminative and robust fusion outputs that better support downstream detection.

4.3.2. Computational efficiency analysis

To further assess the practical applicability of the proposed framework, we compare DAMFusion with the seven competing methods in terms of trainable parameters (Params), floating-point operations (FLOPs), and average inference time. For a fair comparison, all methods are re-implemented and evaluated on the same hardware platform (NVIDIA GeForce RTX 3090 GPU, PyTorch 1.11.0) with paired infrared-visible images of resolution 640×480. FLOPs are estimated on a single forward pass, and the inference time is averaged over 100 randomly sampled image pairs after a 10-iteration warm-up to exclude initialization overhead. For our method, FLOPs and parameter counts are reported for the trainable components; the frozen CLIP-based degradation encoder is excluded because it is used purely as a fixed degradation prior. The results are summarized in Table 5.

As shown in Table 5, the proposed DAMFusion achieves a balanced computational profile across the three metrics, ranking fourth in parameters, FLOPs, and inference time. It is worth noting that the three indicators are not strictly correlated across the eight compared methods, because parameter count, theoretical FLOPs, and wall-clock latency are governed by distinct architectural factors. For instance, UMF has the smallest parameter count and CAMF achieves the lowest FLOPs, yet

Table 5
Comparison of computational efficiency on infrared-visible image pairs of resolution 640×480. The best and second-best results are highlighted in **Bold** and *italic*, respectively.

Method	Params (M)	FLOPs (G)	Time (ms)
CAMF (Tang et al., 2024)	*1.43*	**8.62**	31.7
GIFNet (Cheng et al., 2025)	5.84	*14.27*	**19.4**
SuperFusion (Tang et al., 2022a)	9.65	31.22	38.9
SwinFusion (Ma et al., 2022a)	13.48	27.85	56.3
UMF (Wang et al., 2022)	**0.62**	19.83	*22.6*
DAFusion (Wang et al., 2025)	22.36	58.41	73.5
Text-IF (Yi et al., 2024)	86.71	39.62	49.8
Ours	8.72	22.94	36.4

GIFNet attains the shortest inference time owing to its parallel-friendly multi-task design. Text-IF, despite having by far the largest parameter count due to its semantic text encoder, is not the slowest method, since the text branch can be precomputed and reused once the textual condition is fixed. By contrast, DAFusion has substantially fewer parameters than Text-IF but exhibits the longest runtime because of its sequential degradation-restoration branches, and SwinFusion incurs a relatively high inference time despite a moderate parameter count, reflecting the cost of cascaded self-attention layers. These observations confirm that no single dimension fully characterizes practical efficiency, and a comprehensive comparison must consider all three metrics together. For our method, although the heterogeneous expert design introduces additional total parameters, the Top-K sparse activation mechanism ensures that only a small subset of experts is invoked at inference, which keeps both FLOPs and runtime within a moderate range. As a result, DAMFusion is significantly more efficient than the heaviest degradation-aware baselines (DAFusion and Text-IF), while remaining competitive with lighter general-purpose fusion methods such as GIFNet and SuperFusion. Combined with the consistent quality gains reported in Sections 4.2 and 4.3.1, this indicates that the proposed framework strikes a reasonable trade-off between fusion accuracy, robustness under compound degradations, and computational cost, suggesting good practical applicability in real-world scenarios.

4.4. Ablation experiment

The ablation study evaluates the contribution of each component by progressively removing or replacing key modules. By comparing results across different configurations, we reveal how each factor affects overall performance and provide guidance for optimization. The experimental settings are summarized in Table 6: (a) Without Stage I: The Stage I image restoration is bypassed; degraded images are used directly for fusion training. (b) Without CLIP Model: The CLIP-based image embedding that encodes degradation types is removed, eliminating this guidance for feature extraction and reconstruction. (c) Without KL divergence based mutual information loss: The sparsity regularization introduced by the mutual information loss is omitted, and a standard MoE gating network is used instead. (d) Without MoE: The MoE-based mixture of experts is replaced with a simple CNN, removing the advantages of MoE for han-

Table 6
Ablation experiment. "✓" indicates that the component is included.

	Stage I	CLIP Model	Mutual Information Loss	MoE
(a)		✓	✓	✓
(b)	✓		✓	✓
(c)	✓	✓		✓
(d)	✓	✓	✓	
(e)	✓	✓	✓	✓

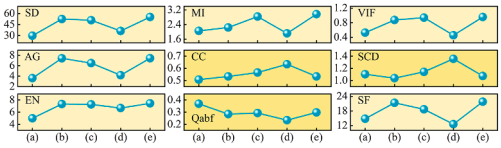


Fig. 12. Ablation experiments.

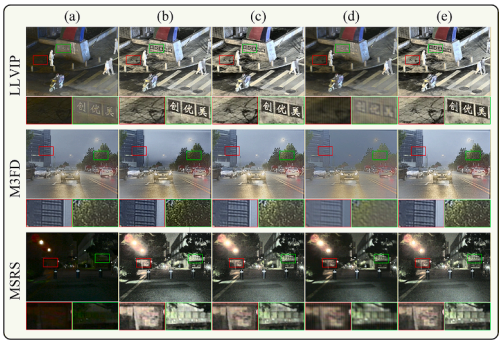


Fig. 13. Visual comparison of ablation studies.

dling complex feature extraction and reconstruction. (e) Full Model: The complete model, with all components enabled for training and evaluation.

All experiments are conducted separately on the M3FD, LLVIP, and MSRS datasets, and we report the average across them to ensure consistent settings and fair evaluation. The results are shown in Fig. 12.

While the complete method is not the top performer on three individual metrics, it remains the strongest overall. Without the sparsity constraint, the model may over-rely on certain experts and over-emphasize specific image features, which boosts SCD and CC. When the CLIP module is removed, the Qabf score becomes slightly higher. A plausible reason is that, without degradation-type guidance, the model focuses more on local edges during feature extraction, thereby raising the edge-preservation metric at the cost of degradation robustness. Overall, the full model excels across multiple metrics, underscoring the importance of component synergy for performance gains. Visual comparisons of the ablations are shown in Fig. 13.

When different components are removed, the quality of the fused images declines to varying degrees, with (a) Without Stage I and (c) Without KL divergence based mutual information loss showing the most pronounced degradation. Specifically, (a) Without Stage I leads to more visible degradation, especially under low-light and blurred conditions, while (c) Without KL divergence based mutual information loss results in blurrier images and insufficient detail restoration. Overall, (e) Full Model delivers the best results, as the synergy of all components effectively enhances the image quality.

5. Conclusion

This paper presents DAMFusion, an end-to-end degradation-aware fusion framework that addresses the limitations of existing multi-modal image fusion methods, particularly in complex degradation scenarios such as noise, low contrast, and low resolution. While current methods often struggle with adaptability, DAMFusion combines autoencoders (AE) and Mixture of Experts (MoE) networks, enhanced by CLIP-based semantic guidance, enabling adaptive feature modeling and fine fusion. The framework incorporates two heterogeneous sparse expert networks (EDMoE and FMoE), using KL divergence constraints and mutual information sparsity control to reduce redundancy and imbalance in expert selection, thus improving performance across different degradation types. Extensive experiments demonstrate that DAMFusion significantly improves image detail restoration, structural integrity, and visual quality, outperforming state-of-the-art methods in infrared-visible and medical image fusion tasks.

However, limitations exist. First, the framework still relies on pseudo-ground truth for degradation modeling, which may reduce robustness in unseen degradation scenarios. Second, the computational complexity introduced by the MoE architecture is moderate, and further reduction would be beneficial for resource-constrained deployment. Future work will focus on enhancing robustness, refining degradation modeling, and improving computational efficiency to further boost the framework's applicability in real-world scenarios.

CRediT authorship contribution statement

Junjie Ma: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Peng Hu:** Supervision, Resources, Project administration, Methodology, Conceptualization; **Yi Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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